

Probability and Stochastics for Finance

A Report based on NPTEL Course by Dr. Joydeep Dutta (IIT Kanpur)

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June 25, 2025

Abstract

This report presents a comprehensive exploration of probability and stochastic processes with a focus on applications in financial mathematics, as taught in the course *NOC: Probability and Stochastics for Finance* by Dr. Joydeep Dutta, IIT Kanpur. Beginning with the foundational axioms of probability, the document builds toward advanced topics including random variables, expectation, convergence theorems, and the law of large numbers. Classical paradoxes and problems such as Buffon's Needle and Bertrand's Paradox are revisited to deepen intuition. Core measure-theoretic tools such as conditional expectation, Radon-Nikodym derivatives, and filtrations are discussed in context. The report then transitions into stochastic processes, covering Brownian motion, random walks, and quadratic variation. Ito calculus, stochastic differential equations, and the Ito formula are developed rigorously. Finally, the application of stochastic modeling in finance is explored through models of interest rates, geometric Brownian motion, and the Black-Scholes framework. Together, these elements provide a strong theoretical and practical foundation for understanding uncertainty and randomness in financial systems.

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1 Probability

1.1 Random Experiments

Definition: A *random experiment* is a physical situation or process that leads to one of several possible outcomes, where the outcome cannot be predicted with certainty in advance, even though the set of all possible outcomes is known.

Assumptions:

- The experiment can be repeated under identical conditions.
- The set of all possible outcomes, known as the *sample space* Ω , is well-defined.
- Each trial of the experiment results in exactly one outcome from Ω .
- There exists an underlying mechanism of chance that governs the outcomes.

Examples of random experiments include tossing a coin, rolling a die, or drawing a card from a shuffled deck.

1.2 Circular Definition of Probability

Definition: The *circular definition of probability* refers to an early approach to defining probability in terms of equally likely outcomes, where the notion of "equally likely" is itself based on intuitive probability, thus creating a circular reasoning loop.

Formally: If a sample space Ω consists of n equally likely outcomes and an event A consists of m of those outcomes, then the probability of A is defined as:

$$P(A) = \frac{m}{n}$$

Assumptions:

- All outcomes in Ω are equally likely.
- The sample space Ω is finite and all outcomes can be enumerated.
- The event A is a well-defined subset of Ω .

Issue: The definition assumes knowledge of what it means for outcomes to be equally likely — which itself presupposes a notion of probability, leading to circularity.

1.3 Kolmogorov Axioms of Probability

Definition: Let there be a sample space Ω , an event space $\mathcal{E}$, and a probability measure P , such that $P(E)$ is the probability of some event $E \in \mathcal{E}$. Then, we introduce three axioms of probability:

1. **First Axiom:** The probability of an event is a non-negative real number:

$$P(E) \in \mathbb{R}, \quad P(E) \geq 0, \quad \text{for all } E \in \mathcal{E}.$$

2. **Second Axiom:** The probability that at least one elementary event in the sample space will occur is one:

$$P(\Omega) = 1.$$

3. **Third Axiom:** The probability of any countable sequence of disjoint (i.e., mutually exclusive) events $E_1, E_2, E_3, \dots$ is equal to the sum of the probabilities of the individual events:

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

1.4 Buffon's Needle Problem

Problem: Given a needle of length l dropped on a plane ruled with parallel lines t units apart, what is the probability that the needle will lie across a line upon landing?

Solution:

Let us define:

- $\theta \in [0, \pi]$: the acute angle between the needle and the parallel lines.
- $x \in [0, t/2]$: the distance from the center of the needle to the nearest line.

The needle crosses a line if the vertical projection of half the needle length is greater than or equal to x , i.e., if:

$$x \leq \frac{\ell}{2} \sin \theta$$

The probability is computed by integrating over all possible positions and orientations of the needle:

$$P = \frac{2}{\pi t} \int_0^{\pi/2} \ell \sin \theta d\theta$$

Evaluating the integral:

$$P = \frac{2\ell}{\pi t} \int_0^{\pi/2} \sin \theta d\theta = \frac{2\ell}{\pi t} [-\cos \theta]_0^{\pi/2} = \frac{2\ell}{\pi t} (1 - 0)$$

$$\boxed{P = \frac{2\ell}{\pi t}}$$

Conclusion: For a needle of length $\ell \leq t$, the probability that it crosses a line upon landing is:

$$\boxed{P = \frac{2\ell}{\pi t}}$$

1.5 Chevalier de Méré's Problem

Problem Statement:

Chevalier de Méré, a 17th-century French nobleman and gambler, posed two famous probability problems:

1. What is the probability of getting at least one six in four rolls of a fair six-sided die?

2. What is the probability of getting at least one double-six in 24 rolls of a pair of fair dice?

He found that the first event usually occurs and bet on it. He also assumed the second event should occur just as frequently, but he started losing money when betting on it.

Solution:

1. At least one six in 4 rolls of a fair die:

The probability of not getting a six in one roll is $\frac{5}{6}$. Hence, the probability of not getting any sixes in 4 rolls is:

$$P(\text{no six in 4 rolls}) = \left(\frac{5}{6}\right)^4$$

$$P(\text{at least one six}) = 1 - \left(\frac{5}{6}\right)^4 = 1 - \frac{625}{1296} \approx 0.5177$$

So, the event occurs more than half the time.

2. At least one double-six in 24 rolls of a pair of dice:

The probability of getting a double-six in one roll of two dice is $\frac{1}{36}$. Thus, the probability of not getting a double-six in one roll is $\frac{35}{36}$, and in 24 rolls:

$$P(\text{no double-six in 24 rolls}) = \left(\frac{35}{36}\right)^{24}$$

$$P(\text{at least one double-six}) = 1 - \left(\frac{35}{36}\right)^{24} \approx 1 - 0.5086 = 0.4914$$

This is less than 0.5, so the event occurs less than half the time.

Conclusion:

Although both events seem intuitively similar, the second one has a lower probability of success. This misconception led de Méré to incorrectly assume he had a favorable bet in both cases. His problems motivated Pascal and Fermat to formalize modern probability theory.

1.6 Bertrand's Paradox

Problem Statement:

Bertrand's Paradox is a famous problem in probability theory that illustrates the ambiguity in defining a "random" chord in a circle. The problem is as follows:

A chord is drawn at random in a circle. What is the probability that its length is greater than the side length of the inscribed equilateral triangle?

Let the radius of the circle be R . The side length of the inscribed equilateral triangle is $\sqrt{3}R$. So, the question reduces to: What is the probability that a randomly drawn chord is longer than $\sqrt{3}R$?

Bertrand showed that the answer depends on how "random" is defined.

Approach 1: Random Endpoints on the Circumference

Select two points independently and uniformly on the circumference, and draw the chord joining them.

The probability that the chord is longer than the side of the inscribed equilateral triangle is:

$$P = \frac{1}{3}$$

Approach 2: Random Radial Distance (Midpoint Method)

Choose a chord by selecting a random point uniformly along a radius and drawing a chord perpendicular to that radius at that point.

Let r be the perpendicular distance from the center to the chord. For the chord to be longer than $\sqrt{3}R$, we require:

$$r < \frac{R}{2}$$

Since r is uniformly distributed between 0 and R , the probability is:

$$P = \frac{1}{2}$$

Approach 3: Random Midpoints Within the Circle

Select a point uniformly at random inside the circle and draw a chord having that point as its midpoint.

Let d be the distance of the midpoint from the center. For the chord to be longer than $\sqrt{3}R$, we require:

$$d < \frac{R}{2}$$

Since the area of a circle grows with the square of the radius, the probability is proportional to the area of the smaller circle of radius $\frac{R}{2}$:

$$P = \frac{\pi \left(\frac{R}{2}\right)^2}{\pi R^2} = \frac{1}{4}$$

Conclusion:

Bertrand's paradox highlights that in geometric probability, the definition of "randomness" or "uniform distribution" is crucial. Different reasonable interpretations can lead to different answers, illustrating the importance of clearly specifying a probability model.

1.7 Definition of Random Variables/Vector

Let $(\Omega, \mathcal{E}, P)$ be a probability space. A **random variable** is a measurable function

$$X : \Omega \rightarrow \mathbb{R}$$

such that for all Borel sets $B \in \mathcal{B}(\mathbb{R})$, the preimage $X^{-1}(B) \in \mathcal{E}$. That is,

$$\forall B \in \mathcal{B}(\mathbb{R}), \quad \{\omega \in \Omega \mid X(\omega) \in B\} \in \mathcal{E}.$$

This ensures that the probability of the event $\{X \in B\}$ is well-defined.

A **random vector** is a measurable function:

$$\mathbf{X} : \Omega \rightarrow \mathbb{R}^n$$

such that each component $X_i : \Omega \rightarrow \mathbb{R}$ is a real-valued random variable for $i = 1, 2, \dots, n$. That is,

$$\mathbf{X}(\omega) = (X_1(\omega), X_2(\omega), \dots, X_n(\omega)).$$

The measurability condition for $\mathbf{X}$ requires that for all Borel sets $B \in \mathcal{B}(\mathbb{R}^n)$, we have:

$$\{\omega \in \Omega \mid \mathbf{X}(\omega) \in B\} \in \mathcal{E}.$$

1.8 Simple Function

A function $f : \Omega \rightarrow \mathbb{R}$ is called a **simple function** if it takes only finitely many values. That is,

$$f(\omega) = \sum_{i=1}^n a_i \mathbb{1}_{A_i}(\omega)$$

where $a_i \in \mathbb{R}$, $A_i \in \mathcal{E}$, and the indicator function $\mathbb{1}_{A_i}(\omega)$ equals 1 if $\omega \in A_i$ and 0 otherwise.

Integration with respect to a probability measure P for a simple function f is defined as:

$$\int_{\Omega} f(\omega) dP(\omega) = \sum_{i=1}^n a_i P(A_i).$$

For a non-negative random variable $X : \Omega \rightarrow \mathbb{R}_{\geq 0}$, the integral is defined as the supremum over integrals of simple functions less than or equal to X :

$$\int_{\Omega} X dP = \sup \left\{ \int_{\Omega} f dP \mid 0 \leq f \leq X, f \text{ is simple} \right\}.$$

Expectation of a random variable X is defined as:

$$\mathbb{E}[X] = \int_{\Omega} X dP,$$

provided the integral exists (i.e., X is integrable). If X is not non-negative, we write:

$$X = X^+ - X^-, \quad \text{where } X^+ = \max(X, 0), \quad X^- = \max(-X, 0),$$

and then define

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-],$$

provided at least one of $\mathbb{E}[X^+]$, $\mathbb{E}[X^-]$ is finite.

1.9 Expectation

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable on the probability space $(\Omega, \mathcal{E}, P)$. The **expectation** or **expected value** of X , denoted by $\mathbb{E}[X]$, is defined as:

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) dP(\omega),$$

provided this integral exists (i.e., X is integrable).

Properties:

- **Linearity:** For any real constants a, b and random variables X, Y :

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

- **Monotonicity:** If $X \leq Y$ almost surely, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.
- **Non-negativity:** If $X \geq 0$ almost surely, then $\mathbb{E}[X] \geq 0$.

For discrete random variables taking values x_i with probabilities p_i , the expectation is:

$$\mathbb{E}[X] = \sum_i x_i p_i.$$

For continuous random variables with density $f_X(x)$, the expectation is:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

1.10 Conditional Probability

Let $A, B \in \mathcal{E}$ with $P(B) > 0$. The **conditional probability** of A given B is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Properties:

- **Bayes' Theorem:** If $\{B_i\}$ is a partition of Ω and $P(B_i) > 0$, then

$$P(B_j | A) = \frac{P(A | B_j)P(B_j)}{\sum_i P(A | B_i)P(B_i)}.$$

- **Law of Total Probability:** If $\{B_i\}$ is a partition of Ω and $P(B_i) > 0$, then

$$P(A) = \sum_i P(A | B_i)P(B_i).$$

1.11 Independent Events

Two events $A, B \in \mathcal{E}$ are said to be **independent** if:

$$P(A \cap B) = P(A)P(B).$$

This definition extends to more than two events. A collection $\{A_1, A_2, \dots, A_n\} \subset \mathcal{E}$ is mutually independent if for every subcollection $\{A_{i_1}, \dots, A_{i_k}\}$, we have:

$$P\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k P(A_{i_j}).$$

Properties:

- If A and B are independent, so are A and B^c , A^c and B , and A^c and B^c .
- Independence implies that knowing one event has occurred does not change the probability of the other.
- Independence $\not\Rightarrow$ mutual exclusivity, and vice versa.

1.12 Chebyshev's Inequality

Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathbb{E}[(X - \mu)^2]$. Then for any $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

This is known as **Chebyshev's inequality**. It provides a bound on the probability that a random variable deviates from its mean by more than a specified amount.

Implication: No matter the distribution, the probability mass is concentrated near the mean as long as the variance is small.

1.13 Borel-Cantelli Lemma

Let $\{A_n\}_{n=1}^\infty$ be a sequence of events in a probability space. Then:

First Borel-Cantelli Lemma: If $\sum_{n=1}^\infty P(A_n) < \infty$, then

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) = 0.$$

The Borel Cantelli Lemma says that if the sum of the probabilities of the $\{A_n\}$ are finite, then the collection of outcomes that occur infinitely often must have probability zero

Second Borel-Cantelli Lemma (if events are independent): If $\{A_n\}$ are independent and $\sum_{n=1}^\infty P(A_n) = \infty$, then

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) = 1.$$

1.14 Almost Sure Convergence

A sequence of random variables $\{X_n\}$ is said to converge **almost surely** (a.s.) to a random variable X if:

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1.$$

This is the strongest form of convergence. It implies that for almost every $\omega \in \Omega$, the values $X_n(\omega)$ eventually get arbitrarily close to $X(\omega)$.

Notation: $X_n \xrightarrow{\text{a.s.}} X$.

1.15 Convergence in Probability

A sequence $\{X_n\}$ converges **in probability** to X if:

$$\forall \varepsilon > 0, \quad \lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0.$$

Notation: $X_n \xrightarrow{P} X$.

Relation to a.s. convergence:

$$X_n \xrightarrow{\text{a.s.}} X \Rightarrow X_n \xrightarrow{P} X,$$

but the converse is not always true.

1.16 Law of Large Numbers and Central Limit Theorem

Weak Law of Large Numbers (WLLN): Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then:

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu.$$

That is, the sample average converges in probability to the true mean.

Strong Law of Large Numbers (SLLN): Under the same assumptions, we also have:

$$\overline{X}_n \xrightarrow{\text{a.s.}} \mu.$$

Central Limit Theorem (CLT): Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 . Then the standardized sum converges in distribution to a standard normal:

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Implication: Even if the underlying distribution is not normal, the sum of a large number of i.i.d. variables behaves like a Gaussian.

2 Conditional Probability

2.1 Conditional Expectation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let X be an integrable random variable ($\mathbb{E}[|X|] < \infty$). Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. The **conditional expectation** of X given $\mathcal{G}$ is the $\mathcal{G}$ -measurable random variable $\mathbb{E}[X | \mathcal{G}]$ such that:

$$\int_G \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}, \quad \text{for all } G \in \mathcal{G}.$$

Properties:

- Linearity: $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$
- Tower property: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$
- If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$

2.2 Partition of Sigma Algebra Expectation

Let $\{A_i\}_{i \in I}$ be a partition of Ω such that each $A_i \in \mathcal{F}$, and $\mathbb{P}(A_i) > 0$. Then the conditional expectation of X given the partition $\{A_i\}$ is defined as:

$$\mathbb{E}[X \mid A_i] = \frac{1}{\mathbb{P}(A_i)} \int_{A_i} X(\omega) d\mathbb{P}(\omega).$$

The full conditional expectation becomes:

$$\mathbb{E}[X \mid \sigma(\{A_i\})](\omega) = \sum_i \mathbb{E}[X \mid A_i] \cdot \mathbf{1}_{A_i}(\omega).$$

2.3 Conditional Expectation When We Do Not Have Partition of Sigma Algebra

Even when there is no partition structure, we can define the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ using the **Radon-Nikodym theorem**. It exists as a $\mathcal{G}$ -measurable function that integrates correctly on $\mathcal{G}$ -measurable sets:

$$\int_G \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}, \quad \forall G \in \mathcal{G}.$$

This definition generalizes to non-discrete spaces and does not require $\mathcal{G}$ to be generated by a partition.

2.4 How to Construct a New Probability Measure from an Old Probability Measure

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $Z : \Omega \rightarrow \mathbb{R}$ be a non-negative random variable with $\mathbb{E}_{\mathbb{P}}[Z] = 1$. Then we can define a new probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ by:

$$\mathbb{Q}(A) = \int_A Z d\mathbb{P}, \quad \forall A \in \mathcal{F}.$$

Z is called the **Radon-Nikodym derivative** of $\mathbb{Q}$ with respect to $\mathbb{P}$:

$$Z = \frac{d\mathbb{Q}}{d\mathbb{P}}.$$

2.5 Radon-Nikodym Theorem

Theorem (Radon-Nikodym): Let $\mathbb{Q}$ be a σ -finite measure absolutely continuous with respect to $\mathbb{P}$ (denoted $\mathbb{Q} \ll \mathbb{P}$). Then there exists a measurable function $Z : \Omega \rightarrow [0, \infty)$ such that for all $A \in \mathcal{F}$:

$$\mathbb{Q}(A) = \int_A Z d\mathbb{P}, \quad \text{and} \quad Z = \frac{d\mathbb{Q}}{d\mathbb{P}}.$$

Z is unique $\mathbb{P}$ -almost surely. This theorem is crucial for defining conditional expectations and changes of measure in probability theory and financial mathematics.

2.6 Existence of Conditional Expectation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $X \in L^1(\mathbb{P})$ be integrable. Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. Then there exists a $\mathcal{G}$ -measurable random variable Y such that:

$$\int_G Y d\mathbb{P} = \int_G X d\mathbb{P} \quad \forall G \in \mathcal{G}.$$

This Y is the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$, and it is unique up to $\mathbb{P}$ -almost sure equality.

Proof Sketch: This follows from the Radon-Nikodym theorem, since the map $G \mapsto \int_G X d\mathbb{P}$ is a finite signed measure on $\mathcal{G}$ absolutely continuous w.r.t. $\mathbb{P}|_{\mathcal{G}}$.

3 Martingales

3.1 Probability Space and Filtration

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a finite probability space, where:

- Ω is the sample space (finite),
- $\mathcal{F}$ is the sigma-algebra of events,
- $\mathbb{P}$ is the probability measure with $\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$.

A filtration $\{\mathcal{D}_k\}_{k=0}^n$ is a sequence of sub- σ -algebras such that:

$$\mathcal{D}_0 \subseteq \mathcal{D}_1 \subseteq \dots \subseteq \mathcal{D}_n \subseteq \mathcal{F},$$

representing the accumulation of information over time.

A stochastic process $\{X_k\}_{k=0}^n$ is said to be **adapted** to the filtration if each X_k is $\mathcal{D}_k$ -measurable.

3.2 Definition

A sequence of integrable random variables $\{X_k\}$ adapted to a filtration/decomposition $\{\mathcal{D}_k\}$ is a **martingale** if:

1. $\mathbb{E}[|X_k|] < \infty$ for all k ,
2. $\mathbb{E}[X_{k+1} \mid \mathcal{D}_k] = X_k$.

3. $\mathbb{E}[|X_k|]$ is $\mathcal{D}_k$ measurable

Submartingale: $\mathbb{E}[X_{k+1} | \mathcal{D}_k] \geq X_k$

Supermartingale: $\mathbb{E}[X_{k+1} | \mathcal{D}_k] \leq X_k$

3.3 Examples of Martingales

1. **Symmetric Random Walk:** Let $\{Z_k\}$ be i.i.d. Bernoulli random variables taking values $+1$ or -1 with equal probability $1/2$. Define:

$$S_k = \sum_{i=1}^k Z_i, \quad \mathcal{D}_k = \sigma(Z_1, \dots, Z_k).$$

Then $(S_k, \mathcal{D}_k)$ is a martingale.

2. **Exponential Martingale:** Let $\{Z_k\}$ be i.i.d. random variables such that $Z_k = \pm 1$ with $\mathbb{P}(Z_k = 1) = p$, $\mathbb{P}(Z_k = -1) = q = 1 - p$. Let:

$$S_k = \sum_{i=1}^k Z_i, \quad X_k = \left(\frac{q}{p}\right)^{S_k}, \quad \mathcal{D}_k = \sigma(Z_1, \dots, Z_k).$$

Then $(X_k, \mathcal{D}_k)$ is a martingale.

3. **Conditional Expectation:** Let N be an integrable random variable and let $\mathcal{D}_k$ be an increasing filtration. Define:

$$E_k = \mathbb{E}[N | \mathcal{D}_k].$$

Then $\{E_k\}$ is a martingale.

4. **Reverse Martingale:** Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables. Define the partial sums:

$$S_k = X_1 + X_2 + \dots + X_k.$$

Now define a sequence of σ -algebras:

$$\mathcal{D}_k = \sigma(S_n, S_{n-1}, \dots, S_k),$$

which represents the information from time k up to n (i.e., "future" information).

Now define the process $\{Y_k\}_k^n$ as:

$$Y_1 = \frac{S_n}{n}, \quad Y_2 = \frac{S_{n-1}}{(n-1)}, \quad \dots, \quad Y_k = \frac{S_{n+1-k}}{(n+1-k)}.$$

Then the process $\{Y_k\}$ is a reverse-time martingale with respect to the decomposition $\{\mathcal{D}_k\}$, i.e.,

$$\mathbb{E}[Y_{k-1} | \mathcal{D}_k] = Y_k.$$

5. **Complex Martingales:** Let $n_1, \dots, n_k$ be independent Bernoulli random variables with $P(n_i = +1) = P(n_i = -1) = \frac{1}{2}$, $S_k = n_1 + \dots + n_k$. Let A and B be integers, $A < 0 < B$. Then with $0 < \lambda < \pi/2$, the sequence $\xi = (\xi_k, \mathcal{F}_k)$ with $\mathcal{F}_k = \mathcal{F}_{S_1}, \dots, S_k$ and

$$\xi_k = (\cos \lambda)^{-k} \exp \left\{ i\lambda \left(S_k - \frac{B+A}{2} \right) \right\}$$

is a complex martingale (i.e., the real and imaginary parts of ξ_k form martingales).

3.4 Properties of Martingales

Let $\{X_k\}$ be a martingale with respect to $\{\mathcal{D}_k\}$. Then:

$$\mathbb{E}[X_k] = \mathbb{E}[X_1] \quad \text{for all } k \geq 1.$$

Additional properties:

- Martingales are preserved under stopping at bounded stopping times.
- Linearity: Sums and scalar multiples of martingales are martingales.

3.5 Stopping Time

A random variable $\tau = \tau(\omega)$ that takes the values $1, 2, \dots, n$ is called a stopping time (with respect to a decomposition $(\mathcal{D}_k)_{1 \leq k \leq n}$, $\mathcal{D}_1 \subseteq \mathcal{D}_2 \subseteq \dots \subseteq \mathcal{D}_n$) if, for $k = 1, \dots, n$, the random variable $I\{\tau = k\}(\omega)$ is $\mathcal{D}_k$ -measurable.

3.6 Optional Stopping Theorem

Let $\xi = (\xi_k, \mathcal{D}_k)_{1 \leq k \leq n}$ be a martingale and τ a stopping time with respect to the decomposition $(\mathcal{D}_k)_{1 \leq k \leq n}$. Then

$$E(\xi_\tau \mid \mathcal{D}_1) = \xi_1,$$

where

$$\xi_\tau = \sum_{k=1}^n \xi_k I_{\{\tau \geq k\}}(\omega)$$

and

$$E\xi_\tau = E\xi_1.$$

3.7 Wald's Identities

For the martingale $(S_k, \mathcal{D}_k)_{1 \leq k \leq n}$ of Example 1, and any stopping time τ (with respect to $(\mathcal{D}_k)$) we have the formulas

$$ES_\tau = 0, \quad ES_\tau^2 = E\tau,$$

known as Wald's identities.

3.8 Ballot Theorem

Let $n_1, \dots, n_n$ be a sequence of independent identically distributed random variables whose values are nonnegative integers, $S_k = n_1 + \dots + n_k$, $1 \leq k \leq n$. Then

$$P\{S_k < k \text{ for all } k, 1 \leq k \leq n \mid S_n\} = \left(1 - \frac{S_n}{n}\right)^+,$$

where $a^+ = \max(a, 0)$.

3.9 Symmetric Random Walk

A symmetric random walk is a discrete-time stochastic process $\{S_n\}_{n \geq 0}$ defined as:

$$S_0 = 0, \quad S_n = \sum_{i=1}^n X_i,$$

where $\{X_i\}$ are i.i.d. random variables with:

$$P(X_i = 1) = P(X_i = -1) = \frac{1}{2}.$$

Each step is independent and equally likely to go up or down. This process is the discrete precursor to Brownian motion.

3.10 Independent Increments

A stochastic process $\{X_t\}_{t \geq 0}$ has **independent increments** if for any $0 \leq t_0 < t_1 < t_2 < \dots < t_n$, the random variables

$$X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \dots, X_{t_n} - X_{t_{n-1}}$$

are independent. This property is key for both Brownian motion and Lévy processes.

3.11 Scaled Random Walk

Let $\{X_i\}$ be i.i.d. with $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = \sigma^2$, and define the scaled random walk:

$$W_n(t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} X_i.$$

As $n \rightarrow \infty$, under Donsker's theorem,

$$W_n(t) \Rightarrow B(t),$$

where $B(t)$ is standard Brownian motion. This provides a bridge between discrete-time random walks and continuous-time Brownian motion.

3.12 Definition of Brownian Motion

A stochastic process $\{B(t)\}_{t \geq 0}$ is called a **standard Brownian motion** (or Wiener process) if it satisfies:

- $B(0) = 0$ almost surely.
- $B(t)$ has independent increments.
- $B(t) - B(s) \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$.
- Paths $t \mapsto B(t)$ are almost surely continuous.

This process forms the foundation of modern stochastic calculus and modeling in finance and physics.

3.13 Filtration Associated with Brownian Motion

Given a Brownian motion $\{B(t)\}_{t \geq 0}$, the natural filtration is:

$$\mathcal{F}_t = \sigma(B(s) : 0 \leq s \leq t),$$

which captures all the information revealed by the Brownian path up to time t .

This filtration satisfies:

- **Adaptedness:** $B(t)$ is $\mathcal{F}_t$ -measurable.
- **Right-continuity:** $\mathcal{F}_t = \bigcap_{\epsilon > 0} \mathcal{F}_{t+\epsilon}$.
- **Usual Conditions:** Completeness and right-continuity.

3.14 Compute Quadratic Variation

Let $\{B(t)\}_{t \geq 0}$ be Brownian motion, and consider a partition $\Pi_n = \{0 = t_0 < t_1 < \dots < t_n = t\}$ with mesh $\|\Pi_n\| \rightarrow 0$. Then the **quadratic variation** over $[0, t]$ is defined as:

$$[B]_t = \lim_{n \rightarrow \infty} \sum_{i=1}^n (B(t_i) - B(t_{i-1}))^2.$$

For Brownian motion, this converges almost surely to:

$$[B]_t = t.$$

3.15 Quadratic Variation of Brownian Motion

For standard Brownian motion $\{B(t)\}$, the quadratic variation over the interval $[0, t]$ is:

$$[B]_t = t.$$

This means that, although Brownian paths are continuous, they are nowhere differentiable and have infinite total variation but finite quadratic variation. This property is foundational in Itô calculus.

Property: If f is a smooth function and $Y_t = f(B_t)$, then Itô's Lemma uses $[B]_t = t$ in its differential form:

$$df(B_t) = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt.$$

4 Ito Integral and Calculus

4.1 Some Important Properties of Itô Integral

The Itô integral $\int_0^t H(s) dB(s)$, where $H(s)$ is an adapted process and $B(s)$ is Brownian motion, satisfies the following key properties:

- **Linearity:** If $a, b \in \mathbb{R}$, and H_1, H_2 are adapted processes,

$$\int_0^t (aH_1(s) + bH_2(s)) dB(s) = a \int_0^t H_1(s) dB(s) + b \int_0^t H_2(s) dB(s).$$

- **Isometry (Itô Isometry):**

$$\mathbb{E} \left[\left(\int_0^t H(s) dB(s) \right)^2 \right] = \mathbb{E} \left[\int_0^t H^2(s) ds \right].$$

- **Martingale Property:** If H is adapted and square-integrable,

$$\int_0^t H(s) dB(s) \text{ is a martingale.}$$

4.2 Simple Processes

A **simple process** is a step function that takes the form:

$$H(t) = \sum_{i=0}^{n-1} H_i \cdot \mathbf{1}_{(t_i, t_{i+1}]}(t),$$

where $\{H_i\}$ are $\mathcal{F}_{t_i}$ -measurable random variables and $0 = t_0 < t_1 < \dots < t_n = T$.

The Itô integral of a simple process H with respect to Brownian motion is defined as:

$$\int_0^T H(s) dB(s) := \sum_{i=0}^{n-1} H_i (B(t_{i+1}) - B(t_i)).$$

4.3 Stochastic Differential Equation (SDE)

A **Stochastic Differential Equation** is an equation of the form:

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t,$$

where $\mu(t, X_t)$ is the drift coefficient, $\sigma(t, X_t)$ is the diffusion coefficient, and B_t is Brownian motion.

The integral form is:

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s.$$

SDEs are used extensively in financial modeling (e.g., Black-Scholes).

4.4 Itô–Doeblin Formula

The Itô–Doeblin formula generalizes the chain rule to stochastic calculus. For a function $f(t, X_t)$ where X_t follows an SDE:

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t,$$

the differential of $f(t, X_t)$ is:

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dB_t.$$

4.5 Quadratic Variation of Itô Process

An **Itô process** is defined as:

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dB_s.$$

Its quadratic variation is given by:

$$[X]_t = \int_0^t \sigma^2(s) ds.$$

This shows that even though X_t is continuous, its path is not differentiable.

4.6 Itô's Formula for Itô Process

Let X_t be an Itô process:

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dB_s,$$

and let $f(t, x) \in C^{1,2}$. Then Itô's formula states:

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu(t) \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2(t) \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma(t) \frac{\partial f}{\partial x} dB_t.$$

This is the cornerstone of stochastic calculus and is used to derive dynamics of functions of stochastic processes.

5 Applications of Ito and Black Scholes

5.1 Higher Dimensional Brownian Motion

A d -dimensional Brownian motion is a vector process

$$\mathbf{B}_t = (B_t^{(1)}, B_t^{(2)}, \dots, B_t^{(d)})$$

where each $B_t^{(i)}$ is an independent standard Brownian motion. The process satisfies:

- $\mathbf{B}_0 = 0$
- $\mathbf{B}_t$ has independent increments
- $\mathbf{B}_t - \mathbf{B}_s \sim \mathcal{N}(0, (t-s)I_d)$ for $0 \leq s < t$
- Paths are continuous almost surely

5.2 Two-Dimensional Itô Formula

Let $\mathbf{X}_t = (X_t^{(1)}, X_t^{(2)})$ be a 2D Itô process:

$$dX_t^{(i)} = \mu^{(i)}(t)dt + \sigma^{(i)}(t)dB_t^{(i)}, \quad i = 1, 2.$$

Let $f(t, x^{(1)}, x^{(2)})$ be twice continuously differentiable. Then:

$$df = \frac{\partial f}{\partial t} dt + \sum_{i=1}^2 \frac{\partial f}{\partial x^{(i)}} dX_t^{(i)} + \frac{1}{2} \sum_{i=1}^2 \frac{\partial^2 f}{\partial (x^{(i)})^2} (dX_t^{(i)})^2.$$

5.3 Lévy's Theorem and Two-Stocks

Lévy's Characterization: A continuous martingale M_t with $M_0 = 0$ and quadratic variation $[M]_t = t$ is a Brownian motion.

In financial markets, let $S_t^{(1)}$ and $S_t^{(2)}$ be two stocks modeled as:

$$dS_t^{(i)} = \mu_i S_t^{(i)} dt + \sigma_i S_t^{(i)} dB_t^{(i)}, \quad i = 1, 2,$$

with correlated Brownian motions:

$$\mathbb{E}[dB_t^{(1)} dB_t^{(2)}] = \rho dt.$$

The joint dynamics are crucial in pricing multi-asset derivatives.

5.4 Geometric Brownian Motion (GBM)

A stochastic process S_t satisfies a GBM if:

$$dS_t = \mu S_t dt + \sigma S_t dB_t.$$

The solution is:

$$S_t = S_0 \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma B_t \right).$$

Used for modeling stock prices under log-normal assumptions.

5.5 Itô Process

An Itô process has the form:

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dB_s,$$

where $\mu(s)$ and $\sigma(s)$ are adapted stochastic processes. This generalizes Brownian motion with drift and stochastic volatility.

5.6 Introduction of Interest Rates

The short-term interest rate r_t is modeled as a stochastic process. The bank account (risk-free asset) evolves as:

$$dB_t = r_t B_t dt.$$

Interest rate models aim to describe the behavior of r_t over time for pricing bonds and interest rate derivatives.

5.7 Vasicek's Model of Interest Rates

The Vasicek model is a mean-reverting model:

$$dr_t = a(b - r_t)dt + \sigma dB_t,$$

where:

- $a > 0$ is the speed of mean reversion,
- b is the long-term mean,
- σ is the volatility.

Solution:

$$r_t = r_0 e^{-at} + b(1 - e^{-at}) + \sigma \int_0^t e^{-a(t-s)} dB_s.$$

5.8 Cox-Ingersoll-Ross Model (CIR Model)

The CIR model is:

$$dr_t = a(b - r_t)dt + \sigma\sqrt{r_t}dB_t.$$

Key property: Ensures $r_t \geq 0$ if $2ab \geq \sigma^2$.

Used to model the evolution of interest rates and default intensities.

5.9 Option in Financial Derivative

An option is a contract giving the holder the right, but not the obligation, to buy (call) or sell (put) an asset at a fixed price before or at a specified expiration date.

European Call Option payoff:

$$C_T = \max(S_T - K, 0).$$

5.10 Assumptions for Black-Scholes Formula

- No arbitrage
- Stock follows GBM: $dS_t = \mu S_t dt + \sigma S_t dB_t$
- Risk-free interest rate r is constant
- No dividends, no transaction costs
- Markets are frictionless and continuous

Under these, the price of a European call is:

$$C(S, t) = S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2),$$

where:

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}, \quad d_2 = d_1 - \sigma\sqrt{T - t}.$$

5.11 Evolution of Portfolio-Value

Let a portfolio hold Δ_t units of stock S_t and the rest in a risk-free asset B_t . The portfolio value:

$$V_t = \Delta_t S_t + \beta_t B_t.$$

Using Itô's lemma and self-financing condition:

$$dV_t = \Delta_t dS_t + \beta_t dB_t.$$

In Black-Scholes framework:

$$dV_t = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \frac{\partial V}{\partial S} dS_t.$$